Lab Practical 10

Design and develop a distributed application to find the coolest/hottest year from the available weather data. Use weather data from the Internet and process it using MapReduce.

Code

Cell 1 — Download Real Weather Data from NASA:

```
import pandas as pd

url = "https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv"
df = pd.read_csv(url, skiprows=1)
df = df[["Year", "J-D"]].copy()
df.columns = ["year", "temperature"]
df = df[df["temperature"] != "***"]
df["temperature"] = df["temperature"].astype(float)
df.to_csv("weather_data.csv", index=False)

print("Downloaded real NASA weather data!")
print(df.head(10))
print(f"\nTotal records: {len(df)}")
```

Cell 2 — Write Mapper:

```
%%writefile mapper.py
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    line = line.strip()
    if line.startswith("year"):
        continue
    try:
        parts = line.split(",")
        year = parts[0].strip()
        temp = float(parts[1].strip())
        print(f"{year}\t{temp}")
    except Exception:
        continue
```

Cell 3 — Write Reducer:

```
%%writefile reducer.py
#!/usr/bin/env python3
import sys

year_temps = {}

for line in sys.stdin:
    line = line.strip()
```

```
    try:
        year, temp = line.split("\t")
        temp = float(temp)
        if year not in year_temps:
            year_temps[year] = []
        year_temps[year].append(temp)
    except Exception:
        continue

avg_temps = {}
for year, temps in year_temps.items():
    avg_temps[year] = sum(temps) / len(temps)

print("\n--- Average Temperature Anomaly Per Year ---")
for year in sorted(avg_temps):
    print(f"Year: {year}  |  Avg Temp Anomaly: {avg_temps[year]:.2f}°C")

coolest_year = min(avg_temps, key=avg_temps.get)
hottest_year = max(avg_temps, key=avg_temps.get)

print("\n--- RESULT ---")
print(f"Coolest Year: {coolest_year}  |  Avg Temp Anomaly: {avg_temps[coolest_year]:.2f}°C")
print(f"Hottest Year: {hottest_year}  |  Avg Temp Anomaly: {avg_temps[hottest_year]:.2f}°C")
```

Output:

```
--- Average Temperature Per Year ---
Year: 1880  |  Avg Temp: -0.17Â°C
Year: 1881  |  Avg Temp: -0.08Â°C
Year: 1882  |  Avg Temp: -0.11Â°C
Year: 1883  |  Avg Temp: -0.17Â°C
Year: 1884  |  Avg Temp: -0.28Â°C
Year: 1885  |  Avg Temp: -0.33Â°C
Year: 1886  |  Avg Temp: -0.31Â°C
Year: 1887  |  Avg Temp: -0.36Â°C
Year: 1888  |  Avg Temp: -0.18Â°C
Year: 1889  |  Avg Temp: -0.11Â°C
Year: 1890  |  Avg Temp: -0.36Â°C
Year: 1891  |  Avg Temp: -0.22Â°C
Year: 1892  |  Avg Temp: -0.27Â°C
Year: 1893  |  Avg Temp: -0.31Â°C
Year: 1894  |  Avg Temp: -0.31Â°C
Year: 1895  |  Avg Temp: -0.23Â°C
Year: 1896  |  Avg Temp: -0.12Â°C
Year: 1897  |  Avg Temp: -0.11Â°C
Year: 1898  |  Avg Temp: -0.28Â°C
Year: 1899  |  Avg Temp: -0.18Â°C
Year: 1900  |  Avg Temp: -0.09Â°C
Year: 1901  |  Avg Temp: -0.16Â°C
Year: 1902  |  Avg Temp: -0.28Â°C
...
--- RESULT ---
Coolest Year: 1909  |  Avg Temp: -0.49Â°C
Hottest Year: 2024  |  Avg Temp: 1.28Â°C
```

Sign: _____.